Research Report (TMO Benchmarking)

Summary:

We benchmarked 17 tone-mapping operators (TMOs) on 992 HDR images. No operator is best everywhere: 15 of them win at least one image, and the best fixed operator (CATMO) wins only 8%. The selector predicts each candidate's TMQI components from 151 published HDR statistics and picks the highest. It finds the exact best operator on 65% of images, but most misses are near ties, so the mean TMQI lost per image is only 0.0033. Whether these differences are visible to people is untested.

Research Problem:

Tone-mapping operators (TMOs) are used to compress high-dynamic-range (HDR) images into 8-bit images that a standard display can show. Perceptual studies agree that no single operator is best for every image (Ledda et al. 2005; Čadík et al. 2008; Cerdá-Company et al. 2018). The ranking depends on the scene and on the quality criterion.

The benchmark shows the same. Across 992 HDR images and 17 operators:

- 15 different operators are the best choice on at least one image. Two TMOs (Pattanaik and van Hateren) never won a single case; as such, they were discarded from the selector.
- The most frequent winner (ATT; Khan et al. 2018) is best on only 25% of images, yet ATT ranks 9th–17th on 20% of images.
- The operator with the best average score (CATMO; Li et al. 2024) is best on only 8% of images, yet CATMO is in the top 5 on 73% of images and never ranks below 12th.

Always applying one fixed operator, even the best one on average, leaves quality unused on most images. Choosing the best operator per image has a cost. It means running every operator and scoring every output against the HDR reference with a metric such as TMQI. This multiplies the cost of tone mapping by the number of candidates.

Therefore, a tone-mapping selector that looks only at the HDR input could predict which operator will score best on each image, without running every operator or computing any metric. This is a case of per-instance algorithm selection (Rice 1976; Kerschke et al. 2019).

We consider three parts for the tone-mapping selector:

1. Quality. How much of the gap between the best fixed operator and per-image best does the selector recover?
2. Accuracy vs quality. How often does it pick the best operator and what does the wrong pick cost?
3. Cost. How long does selection take per image, compared with running and scoring every candidate?

Methodology:

Data:

Byte-identical copies and near-duplicates were removed from the HDR-Real and HDR-Synth sets (Liu et al. 2020), and 1000 unique HDR images were then drawn from what remained. HDR-Synth gathers public collections, including those of Fairchild (2007), Funt & Shi (2010) and Xiao et al. (2002). Every image was resized so that the long side is 512px. On 8 images at least one operator (Ashikhmin, Kim–Kautz or van Hateren) failed, either producing a degenerate output or crashing, so these were discarded. The remaining 992 images were tone-mapped successfully by every operator; these form the analysis set.

Operators:

17 TMOs were run with their default parameters: 13 classical operators through Luminance HDR, ATT, and three learned operators (CATMO, UnCLTMO, and Unpaired-HDR). (Table 1)

operator	reference	type	implementation
Reinhard 2002†	Reinhard et al. 2002	classical	luminance-hdr
Reinhard–Devlin†	Reinhard & Devlin 2005	classical	luminance-hdr
Drago†	Drago et al. 2003	classical	luminance-hdr
Durand	Durand & Dorsey 2002	classical	luminance-hdr
Fattal†	Fattal et al. 2002	classical	luminance-hdr
Mantiuk 2006	Mantiuk et al. 2006	classical	luminance-hdr
Mantiuk 2008	Mantiuk et al. 2008	classical	luminance-hdr
Pattanaik†	Pattanaik et al. 2000	classical	luminance-hdr
Ashikhmin†	Ashikhmin 2002	classical	luminance-hdr
Mai	Mai et al. 2011	classical	luminance-hdr
Lischinski†	Lischinski et al. 2006	classical	luminance-hdr
van Hateren	van Hateren 2006	classical	luminance-hdr
Kim–Kautz†	Kim & Kautz 2008	classical	luminance-hdr
ATT	Khan et al. 2018	classical	Python port of MATLAB code
CATMO	Li et al. 2024	learned	published weights
UnCLTMO	Cao et al. 2024	learned	published weights
Unpaired-HDR	Vinker et al. 2021	learned	published weights

Table 1. The 17 tone-mapping operators, run with their default parameters. † These Luminance HDR operators return linear values, so an output gamma of 2.2 was applied (-G 2.2); without it they score far too low.

Quality metric and labels:

Every output was scored with the Tone-Mapped image Quality Index (TMQI; Yeganeh & Wang 2013) against the original HDR. Each image's label is the operator with the highest TMQI, and the 15 operators that won at least one image are the selector's candidates. TMQI combines two terms:

- Structural fidelity **S**, a multi-scale measure, adapted from SSIM (Wang et al. 2004), that compares the output's local structure with the HDR's.
- Statistical naturalness **N**, which depends on the output's mean brightness and its mean local contrast (over 11×11 blocks).

$$TMQI = a \cdot S^{\alpha} + (1 - a) \cdot N^{\beta}, \text{ with } a = 0.8012, \alpha = 0.3046, \beta = 0.7088.$$

Two other metrics were also computed: FSITM (Nafchi et al. 2015) and TMQI-II (Ma et al. 2015). These are used in the Robustness section.

Features:

The selector sees only the HDR image. It uses 151 features from published sources in three families: HDR image statistics, the statistics each operator computes from its input before mapping, and statistics used by image-quality models. More details are outlined below. (Table 2)

category	source	features	what it measures
HDR image statistics	Pouli et al. 2010	27	log-luminance moments, power-spectrum slope, gradient and Laplacian-band statistics, Lαβ colour
HDR image statistics	Hulusic et al. 2017	5	perceived dynamic range, image key, bright area, PU21 contrast, colourfulness
operator set-up statistics	Reinhard et al. 2002; Reinhard 2002	5	log-average luminance, default white point, automatic key and white-point estimates
operator set-up statistics	Drago et al. 2003	2	max / log-average ratio, log-base divider
operator set-up statistics	Durand & Dorsey 2002	4	bilateral base-layer range, compression factor, detail energy
operator set-up statistics	Fattal et al. 2002	6	average gradient and attenuation at the attenuated pyramid levels
operator set-up statistics	Reinhard & Devlin 2005	2	image key k, contrast m
operator set-up statistics	Kim & Kautz 2008	2	log-mean, global compression k1
operator set-up statistics	Mai et al. 2011	2	log-histogram normaliser Σp^(1/3), largest tone-curve slope share
operator set-up statistics	Mantiuk et al. 2008	12	band-contrast distribution per pyramid level, conditional on background
quality-model statistics	Yeganeh & Wang 2013	10	TMQI's HDR-side local-σ visibility at its 5 scales

category	source	features	what it measures
quality-model statistics	Mittal et al. 2012 (Kundu et al. 2017)	36	BRISQUE natural-scene statistics of log luminance, 2 scales
quality-model statistics	ITU-T P.910	1	spatial information of log luminance
quality-model statistics	Yeganeh & Wang 2013 (adapted)	37	11 / 22 / 44-px block statistics of log luminance
total		**151**	

Table 2. The 151 selector features are in 14 groups, each following one published source. All are computed from the HDR input alone.

Selector:

The selector consists of 45 LightGBM regressors (Ke et al. 2017), 3 for each of the 15 winning TMOs, each one predicting a different component of the TMQI score:

- The structural fidelity S.
- The output's mean brightness.
- The output's mean local contrast.

Each regressor is bagged five times (Breiman 1996), so 15 × 3 × 5 = 225 models run for each prediction. The last two values make up the naturalness term N. TMQI is then computed using the known formula, and the operator with the highest predicted score is chosen. If no operator is predicted to beat CATMO, the selector keeps CATMO.

Predicting components of TMQI rather than classifying the winner directly lets the model reuse the known formula instead of having to learn it. It also matters because S and N behave differently across operators.

Evaluation:

We obtained all results from out-of-fold predictions in 5-fold cross-validation (scikit-learn; Pedregosa et al. 2011), so every image is scored by a model that never saw it. Folds are grouped by scene; after duplicate removal, each image is its own scene. The selector is compared with two reference points: always using CATMO (the floor) and the per-image best operator (the ceiling). The measures are mean TMQI, gap recovered (floor to ceiling), exact accuracy, the share of images within 0.005 TMQI of the best, and regret (TMQI lost compared to the ceiling).

Confidence intervals are 95% bootstrap intervals (Efron & Tibshirani 1993). Paired comparisons use Wilcoxon signed-rank tests (Wilcoxon 1945) with Holm correction (Holm 1979). The feature set and the model were chosen on the same images they are scored on, so the whole choice was repeated inside nested cross-validation (Varma & Simon 2006).

Timing was measured on 50 randomly drawn images on one Apple M3 Pro laptop, one image at a time. Every operator runs exactly as in the benchmark: each Luminance HDR operator is one Docker call, each learned operator runs on the GPU with model loading included, and ATT runs in-process. Exhaustive search runs all 15

candidates and scores each output with TMQI; the selector computes the features, runs the models, and runs only the chosen operator.

Results:

The benchmark sweep shows that operators differ mostly in naturalness: S ranges from 0.77 to 0.91 across the 15 candidates, while N ranges from 0.30 to 0.80. The best fixed operator also depends on the image source and on the metric. On HDR-Real, ATT (0.9306) edges out CATMO (0.9303), and under the FSITM–TMQI average and TMQI-II the best fixed operator is Reinhard 2002 (Table 7). The table below shows how every TMO scored on the dataset sweep. (Table 3)

Operator	Type	Mean TMQI	S	N	Best on	Top 5 on	Bottom half on	Mean rank	Time (s)
CATMO	learned	0.9357	0.873	0.795	81 (8%)	73%	5%	4.3	2.67
Reinhard 2002	classical	0.9326	0.883	0.761	195 (20%)	66%	10%	4.4	0.35
ATT	classical	0.9304	0.894	0.733	249 (25%)	65%	20%	5.0	0.05
Lischinski	classical	0.9217	0.885	0.692	67 (7%)	52%	14%	5.5	0.40
Mai	classical	0.9003	0.877	0.581	55 (6%)	33%	46%	7.8	0.34
UnCLTMO	learned	0.8967	0.872	0.568	43 (4%)	27%	44%	8.0	4.86
Drago	classical	0.8967	0.865	0.580	48 (5%)	27%	52%	8.4	0.39
Mantiuk 2008	classical	0.8958	0.903	0.510	39 (4%)	27%	57%	8.3	0.38
Kim–Kautz	classical	0.8948	0.871	0.562	29 (3%)	29%	46%	8.0	0.37
Durand	classical	0.8836	0.867	0.510	24 (2%)	21%	55%	9.0	0.40
Unpaired-HDR	learned	0.8792	0.854	0.503	34 (3%)	17%	66%	9.9	3.90
Mantiuk 2006	classical	0.8788	0.907	0.416	47 (5%)	22%	66%	9.6	0.52
Fattal	classical	0.8728	0.904	0.403	36 (4%)	23%	58%	9.6	1.18
Reinhard–Devlin	classical	0.8313	0.828	0.304	37 (4%)	9%	84%	12.4	0.36
Ashikhmin	classical	0.8280	0.769	0.376	8 (1%)	9%	83%	12.5	0.45
Pattanaik	classical	0.8155	0.856	0.171	0 (0%)	1%	97%	14.0	not timed
van Hateren	classical	0.7284	0.688	0.080	0 (0%)	1%	98%	16.2	not timed

Table 3. Every operator under TMQI over 992 images, sorted by mean TMQI. S = structural fidelity, N = statistical naturalness. Rank is among all 17 operators on each image; bottom half = ranked 9th–17th. Time = mean seconds per image over the 50 timed images. The three best operators on average are within 0.006 TMQI of each other, yet they win very different numbers of images.

TMQI Improvement:

The selector raises mean TMQI from 0.9357 (always CATMO) to 0.9596, a gain of 0.0239 (95% CI 0.0222 to 0.0257). The ceiling is 0.9629, so the selector recovers 88% of the gap between the floor and the ceiling. It scores higher than CATMO on 835 images and lower on 98. (Table 4, Fig. 1)

It also beats two simpler alternatives. One is the same selector using only the 27 features of Pouli et al. 2010 (0.9547). The other is the project's earlier random-forest classifier (Breiman 2001) with 21 hand-made features (0.9554).

method	**mean TMQI**	**gap recovered**	**exact accuracy**	**within 0.005 of best**
random operator (expected)	0.8919	−161%	6.7%	10.7%
always CATMO (best fixed operator)	0.9357	0%	8.2%	18.1%
selector, Pouli et al. 2010 features only (27)	0.9547	70%	50.7%	64.9%
earlier random-forest classifier (21 hand-made features)	0.9554	73%	52.2%	67.6%
final selector (151 features, structured LightGBM)	**0.9596**	**88%**	**65.2%**	**80.4%**
ceiling (best operator per image)	0.9629	100%	100%	100%

Table 4. Mean TMQI of the chosen operator over 992 images. Gap recovered = (method − always CATMO) / (ceiling − always CATMO).

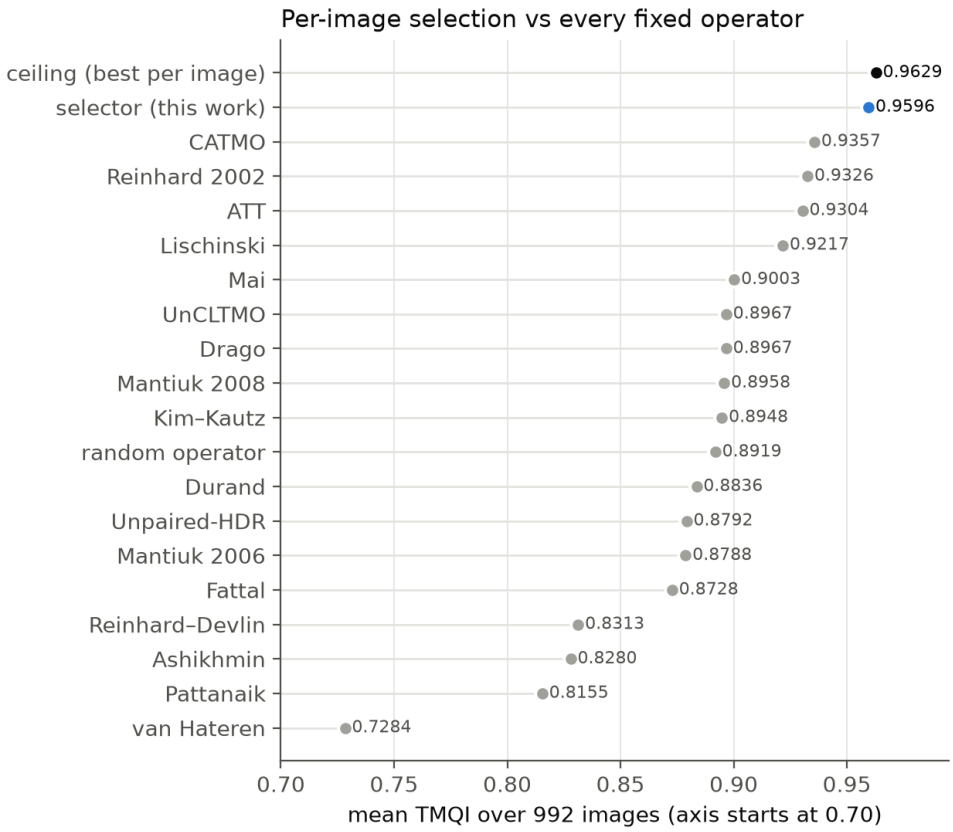


Figure 1. Mean TMQI over 992 images of every fixed operator, a random operator, the selector and the ceiling (best operator per image).

The gain holds in both image sets. On HDR-Real, mean TMQI goes from 0.9303 to 0.9548, against a ceiling of 0.9580. On HDR-Synth, it goes from 0.9449 to 0.9677, against a ceiling of 0.9711. (Table 5)

source	images	always CATMO	selector	ceiling	gap recovered	exact accuracy
HDR-Real	626	0.9303	0.9548	0.9580	88%	64.4%
HDR-Synth	366	0.9449	0.9677	0.9711	87%	66.7%
all	**992**	**0.9357**	**0.9596**	**0.9629**	**88%**	**65.2%**

Table 5. Mean TMQI by image source.

Accuracy vs Quality:

The selector picks the exact best operator on 65% of images. This looks low, but exact accuracy counts every miss as a failure, however small. On many images, the top operators score almost the same. The median difference between the best and second-best operator is 0.0062, and the two are within 0.005 of each other on 44% of images. Most misses are near misses. On the 345 images where the selector misses, it picks the second-best operator 58% of the time and one of the top three 79% of the time. (Fig. 2, Fig. 3)

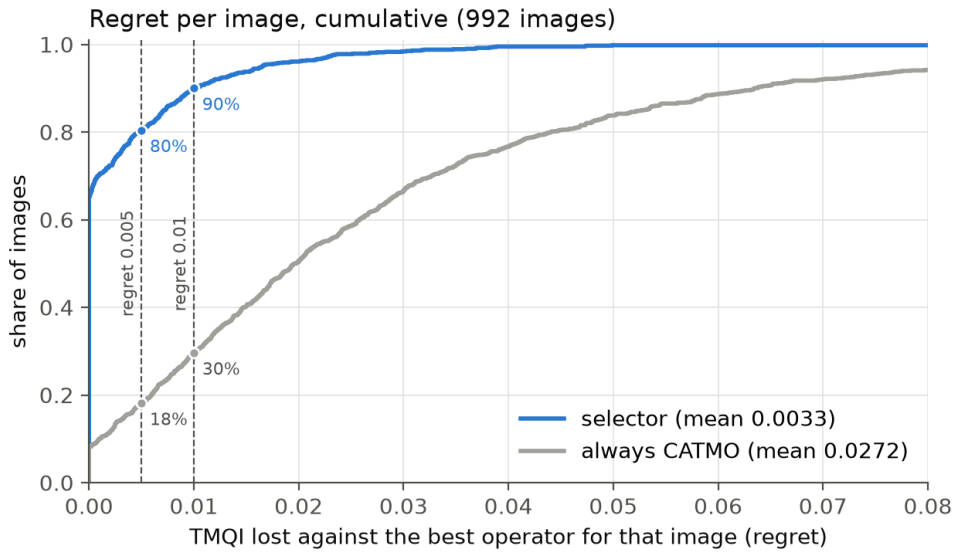


Figure 2. Cumulative distribution of regret (TMQI lost against the best operator for each image), selector vs always CATMO. Dots mark the share of images within 0.005 and 0.01 of the best.

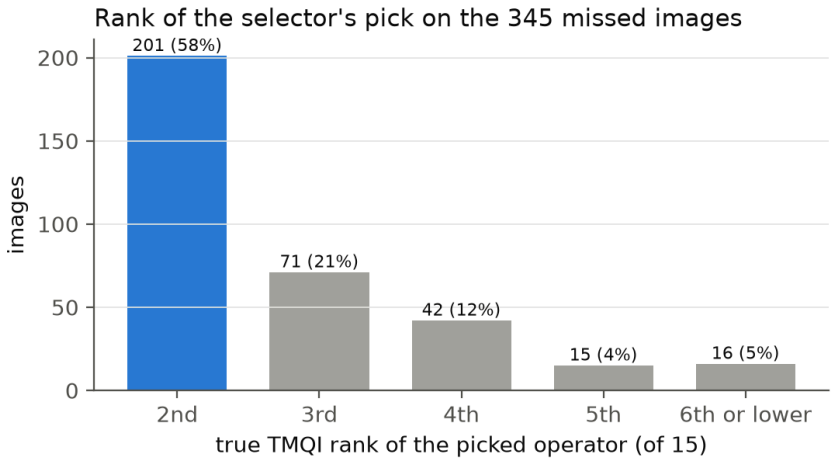


Figure 3. True TMQI rank of the operator the selector picked, on the 345 images where it did not pick the best one.

Computation Time:

The selector takes 1.09 seconds per image on average, against 17.5 seconds for an exhaustive search. Comparing these mean times, it is 16 times faster. Computing the features and running the models takes only 0.17 s; the selector spends most of its time running the chosen operator.

Exhaustive search spends a lot of time running all 15 operators (16.6 s). Scoring the 15 outputs with TMQI adds another 0.92 s. The selector's time depends on what it picks. When it picks a classical operator or ATT, it takes 0.48 s on average. When it picks a learned operator, it takes 3.5 s. (Table 6, Fig. 4)

The 1.09 s mean is pessimistic. Learned operators were picked on 10 of the 50 timed images (20%), but on only 12% of all 992 images. Weighting the two cases by

how often they occur across all 992 images gives about 0.86 s per image, about 20 times faster than exhaustive search.

step	selector	exhaustive search
HDR features (151)	0.12 (0.13)	—
model inference (15 × 3 × 5 = 225 LightGBM models)	0.05 (0.05)	—
tone mapping	0.92 (0.35), 1 operator	16.6 (16.6), 15 operators
TMQI scoring	—	0.92 (0.95), 15 outputs
total	**1.09 (0.52)**	**17.5 (17.5)**
selector total, classical or ATT pick (40 images)	0.48 (range 0.16–1.46)	
selector total, learned-operator pick (10 images)	3.52 (range 2.73–4.47)	

Table 6. Wall-clock seconds per image: mean over 50 images (median in brackets).

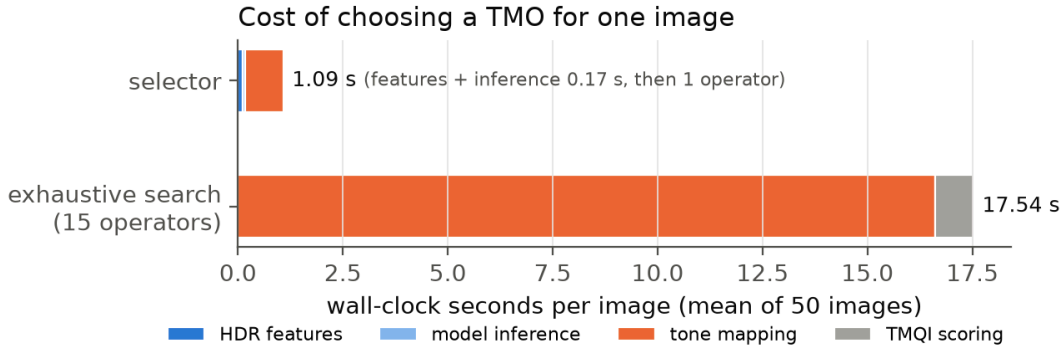


Figure 4. Wall-clock time per image, by step: selector vs exhaustive search over the 15 candidate operators.

Robustness:

Nested cross-validation chose the same configuration in all five folds and reproduced the same result (0.9596). Across five random splits, the selector averages 0.9597 ± 0.00015. The same method, trained for other metrics, also beats the best fixed operator: it recovers 86% of the gap under the FSITM–TMQI average and 58% under TMQI-II. (Table 7)

Metric	Best fixed operator	Selector	Ceiling	Gap recovered	Exact accuracy
TMQI (reference)	0.9357 (CATMO)	0.9596	0.9629	88%	65.2%
FSITM–TMQI average	0.9196 (Reinhard 2002)	0.9309	0.9328	86%	67.7%
TMQI-II	0.8951 (Reinhard 2002)	0.9142	0.9281	58%	61.1%

Table 7. The same method trained and scored under each metric. The floor is the best fixed operator under that metric.

Limitations:

The main limitation is that every gain is measured by TMQI, the metric the selector was trained to maximise.

1. It is not known whether TMQI differences of about 0.005 are noticeable, and TMQI and TMQI-II pick the same best operator on only 11% of images.
2. All 992 images were scaled to 512 px on the long side and collected from two sources. All operators were used on default settings.
3. About 4% of images lose more than 0.02 TMQI against the best operator.

Conclusion:

A selector that looks only at the HDR input recovers 88% of the TMQI available from choosing the best operator per image. It raises mean TMQI from 0.9357 to 0.9596, close to the ceiling of 0.9629, and it takes 1.09 s per image instead of 17.5 s.

Its exact accuracy of 65% understates its quality. Most misses pick the second-best operator, which is usually within 0.006 TMQI of the best, so the average loss against the ceiling is only 0.0033.

The main open question is whether people can see these differences. TMQI is a proxy for human judgement, and it is not known whether differences of about 0.005 are noticeable, so a subjective study is the natural next step.

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Software. Luminance HDR 2.6.1.1 (github.com/LuminanceHDR/LuminanceHDR); tmqi-revised, a Python port of TMQI (github.com/dvolgyes/TMQI); LightGBM 4.6.0; scikit-learn 1.7.2. Operator code: CATMO, UnCLTMO, Unpaired-HDR; ATT is our Python port of the authors' MATLAB code.